

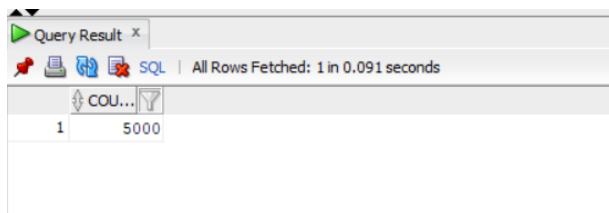
Supply Chain Analytics - Demand Forecasting & Anomaly Detection

Import the Cleaned Excel File into SQL

Verify the Data---SELECT * FROM SUPPLY_CHAIN;

Check the Number of Records

SELECT COUNT(*) FROM SUPPLY_CHAIN;

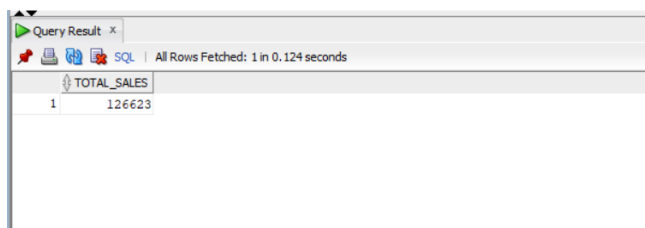


A screenshot of a SQL query result window. The title bar says 'Query Result x'. Below the title bar, there are icons for a red flag, a document, a database, and a SQL icon, followed by the text 'All Rows Fetched: 1 in 0.091 seconds'. The query result is displayed in a table with one column labeled 'COUNT(*)' and one row with the value '5000'.

COUNT(*)
5000

Total Sales Quantity

SELECT SUM(SALES_QUANTITY) AS TOTAL_SALES FROM SUPPLY_CHAIN;

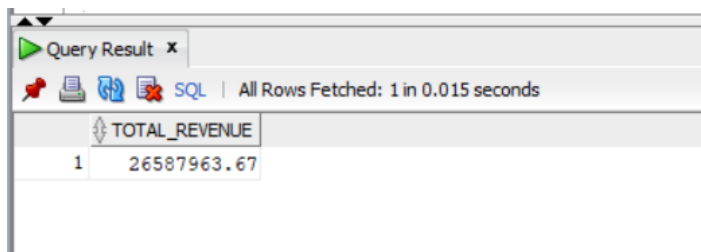


A screenshot of a SQL query result window. The title bar says 'Query Result x'. Below the title bar, there are icons for a red flag, a document, a database, and a SQL icon, followed by the text 'All Rows Fetched: 1 in 0.124 seconds'. The query result is displayed in a table with one column labeled 'TOTAL_SALES' and one row with the value '126623'.

TOTAL_SALES
126623

Total Revenue

SELECT SUM(REVENUE) AS TOTAL_REVENUE FROM SUPPLY_CHAIN;

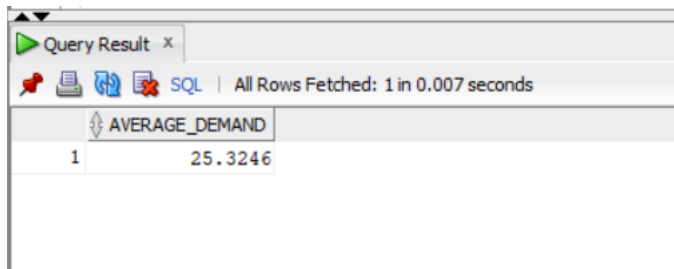


A screenshot of a SQL query result window. The title bar says 'Query Result x'. Below the title bar, there are icons for a red flag, a document, a database, and a SQL icon, followed by the text 'All Rows Fetched: 1 in 0.015 seconds'. The query result is displayed in a table with one column labeled 'TOTAL_REVENUE' and one row with the value '26587963.67'.

TOTAL_REVENUE
26587963.67

Average Demand

```
SELECT AVG(SALES_QUANTITY) AS AVERAGE_DEMAND FROM SUPPLY_CHAIN;
```

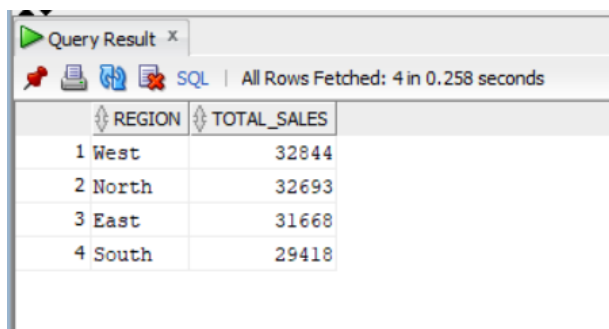


The screenshot shows a 'Query Result' window with a single column 'AVERAGE_DEMAND' and one row with the value 25.3246. The status bar indicates 'All Rows Fetched: 1 in 0.007 seconds'.

	AVERAGE_DEMAND
1	25.3246

Region-wise Sales

```
SELECT  
    REGION,  
    SUM(SALES_QUANTITY) AS TOTAL_SALES  
FROM SUPPLY_CHAIN  
GROUP BY REGION  
ORDER BY TOTAL_SALES DESC;
```

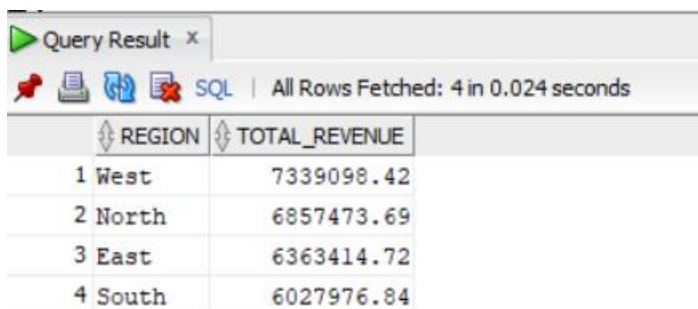


The screenshot shows a 'Query Result' window with two columns: 'REGION' and 'TOTAL_SALES'. It lists four regions: West (32844), North (32693), East (31668), and South (29418). The status bar indicates 'All Rows Fetched: 4 in 0.258 seconds'.

	REGION	TOTAL_SALES
1	West	32844
2	North	32693
3	East	31668
4	South	29418

Region-wise Revenue

```
SELECT REGION, SUM(REVENUE) AS TOTAL_REVENUE FROM  
SUPPLY_CHAIN GROUP BY REGION ORDER BY TOTAL_REVENUE  
DESC;
```

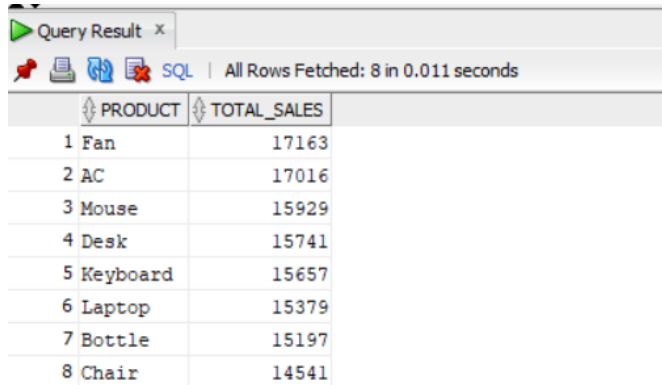


The screenshot shows a 'Query Result' window with two columns: 'REGION' and 'TOTAL_REVENUE'. It lists four regions with their respective revenues: West (7339098.42), North (6857473.69), East (6363414.72), and South (6027976.84). The status bar indicates 'All Rows Fetched: 4 in 0.024 seconds'.

	REGION	TOTAL_REVENUE
1	West	7339098.42
2	North	6857473.69
3	East	6363414.72
4	South	6027976.84

Product-wise Sales

```
SELECT PRODUCT, SUM(SALES_QUANTITY) AS TOTAL_SALES  
FROM SUPPLY_CHAIN GROUP BY PRODUCT ORDER BY  
TOTAL_SALES DESC;
```



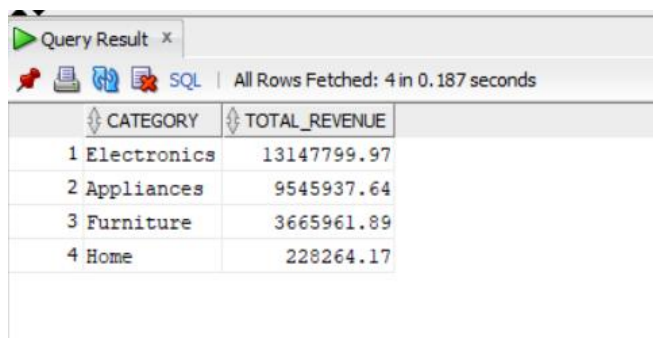
Query Result x

All Rows Fetched: 8 in 0.011 seconds

	PRODUCT	TOTAL_SALES
1	Fan	17163
2	AC	17016
3	Mouse	15929
4	Desk	15741
5	Keyboard	15657
6	Laptop	15379
7	Bottle	15197
8	Chair	14541

Category-wise Revenue

```
SELECT CATEGORY, SUM(REVENUE) AS TOTAL_REVENUE FROM  
SUPPLY_CHAIN GROUP BY CATEGORY ORDER BY TOTAL_REVENUE  
DESC;
```



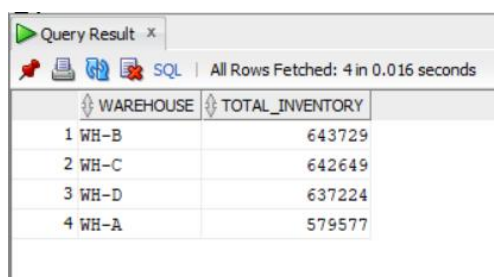
Query Result x

All Rows Fetched: 4 in 0.187 seconds

	CATEGORY	TOTAL_REVENUE
1	Electronics	13147799.97
2	Appliances	9545937.64
3	Furniture	3665961.89
4	Home	228264.17

Warehouse Inventory

```
SELECT WAREHOUSE, SUM(INVENTORY) AS TOTAL_INVENTORY  
FROM SUPPLY_CHAIN GROUP BY WAREHOUSE ORDER BY  
TOTAL_INVENTORY DESC;
```



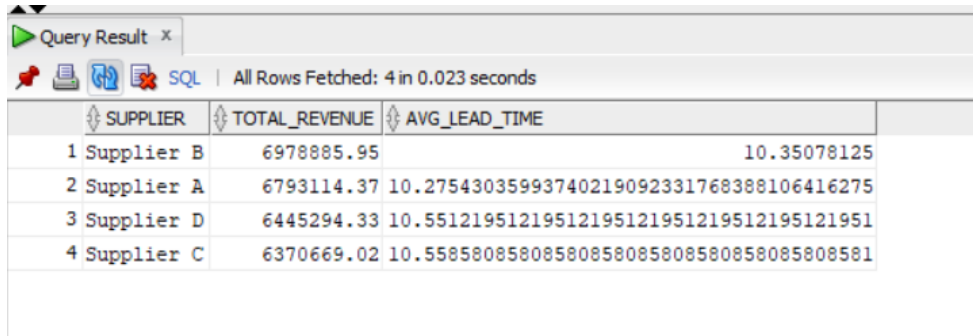
Query Result x

All Rows Fetched: 4 in 0.016 seconds

	WAREHOUSE	TOTAL_INVENTORY
1	WH-B	643729
2	WH-C	642649
3	WH-D	637224
4	WH-A	579577

Supplier Performance

```
SELECT SUPPLIER, SUM(REVENUE) AS TOTAL_REVENUE,  
AVG(LEAD_TIME_DAYS) AS AVG_LEAD_TIME FROM SUPPLY_CHAIN  
GROUP BY SUPPLIER ORDER BY TOTAL_REVENUE DESC;
```

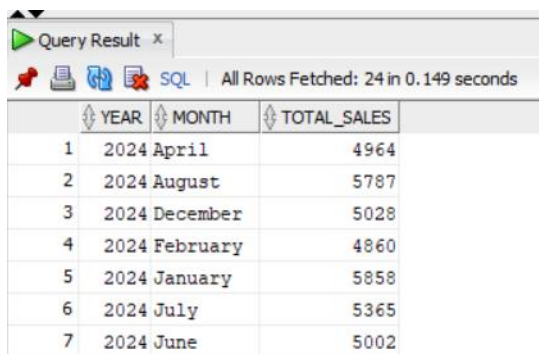


The screenshot shows a SQL query result window with the title 'Query Result x'. It displays the results of a query that ranks suppliers by total revenue. The window includes a toolbar with icons for saving, refreshing, and executing queries, along with a status bar indicating 'All Rows Fetched: 4 in 0.023 seconds'. The data is presented in a table with three columns: SUPPLIER, TOTAL_REVENUE, and AVG_LEAD_TIME. The results are ordered by total revenue in descending order.

	SUPPLIER	TOTAL_REVENUE	AVG_LEAD_TIME
1	Supplier B	6978885.95	10.35078125
2	Supplier A	6793114.37	10.27543035993740219092331768388106416275
3	Supplier D	6445294.33	10.551219512195121951219512195121951
4	Supplier C	6370669.02	10.5585808580858085808580858085808581

Monthly Sales

```
SELECT YEAR,MONTH, SUM(SALES_QUANTITY) AS TOTAL_SALES  
FROM SUPPLY_CHAIN GROUP BY YEAR, MONTH ORDER BY YEAR,  
MONTH;
```



The screenshot shows a SQL query result window with the title 'Query Result x'. It displays the results of a query that aggregates sales data by year and month. The window includes a toolbar with icons for saving, refreshing, and executing queries, along with a status bar indicating 'All Rows Fetched: 24 in 0.149 seconds'. The data is presented in a table with three columns: YEAR, MONTH, and TOTAL_SALES. The results are ordered by year and then by month.

	YEAR	MONTH	TOTAL_SALES
1	2024	April	4964
2	2024	August	5787
3	2024	December	5028
4	2024	February	4860
5	2024	January	5858
6	2024	July	5365
7	2024	June	5002

Top 5 Products

```
SELECT * FROM (  
    SELECT PRODUCT,  
        SUM(REVENUE) AS TOTAL_REVENUE,  
        DENSE_RANK() OVER (ORDER BY SUM(REVENUE) DESC) AS  
        RNK  
    FROM SUPPLY_CHAIN  
    GROUP BY PRODUCT  
)  
WHERE RNK <= 5;
```

Query Result x			
All Rows Fetched: 5 in 0.169 seconds			
	PRODUCT	TOTAL_REVENUE	RNK
1	Laptop	12280383.44	1
2	AC	8517843.93	2
3	Desk	2357659.91	3
4	Chair	1308301.98	4
5	Fan	1028093.71	5

Low Inventory Products

SELECT PRODUCT, INVENTORY FROM SUPPLY_CHAIN WHERE INVENTORY < 100;

Query Result x		
Fetched 50 rows in 0.045 seconds		
	PRODUCT	INVENTORY
1	Chair	19
2	Keyboard	1
3	Keyboard	18
4	Bottle	22
5	Laptop	50
6	Bottle	65
7	Desk	24

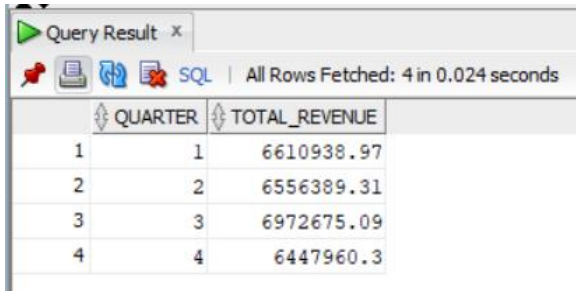
Average Lead Time by Supplier

SELECT SUPPLIER, AVG(LEAD_TIME_DAYS) AS AVG_LEAD_TIME FROM SUPPLY_CHAIN GROUP BY SUPPLIER;

Query Result x	
All Rows Fetched: 4 in 0.016 seconds	
	SUPPLIER
	AVG_LEAD_TIME
1	Supplier D
2	Supplier A
3	Supplier B
4	Supplier C

Revenue by Quarter

SELECT QUARTER, SUM(REVENUE) AS TOTAL_REVENUE FROM SUPPLY_CHAIN GROUP BY QUARTER ORDER BY QUARTER;



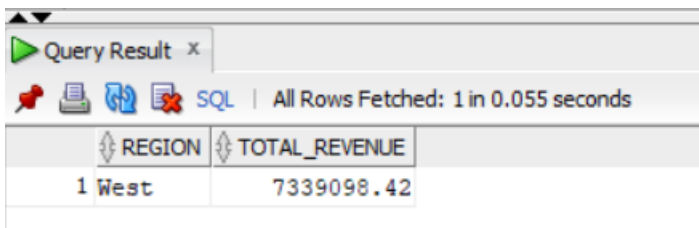
	QUARTER	TOTAL_REVENUE
1	1	6610938.97
2	2	6556389.31
3	3	6972675.09
4	4	6447960.3

Highest Revenue Region

```

SELECT region,
       total_revenue
FROM (
    SELECT region,
           SUM(revenue) AS total_revenue,
           DENSE_RANK() OVER (
               ORDER BY SUM(revenue) DESC
           ) AS rnk
    FROM supply_chain
    GROUP BY region
)
WHERE rnk = 1;

```



	REGION	TOTAL_REVENUE
1	West	7339098.42

Lowest Inventory Warehouse

```

SELECT warehouse,
       total_inventory
FROM (
    SELECT warehouse,
           SUM(inventory) AS total_inventory,

```

```

DENSE_RANK() OVER (
    ORDER BY SUM(inventory)
) AS rnk
FROM supply_chain
GROUP BY warehouse
)
WHERE rnk = 1;

```



Query Result x

All Rows Fetched: 1 in 0.008 seconds

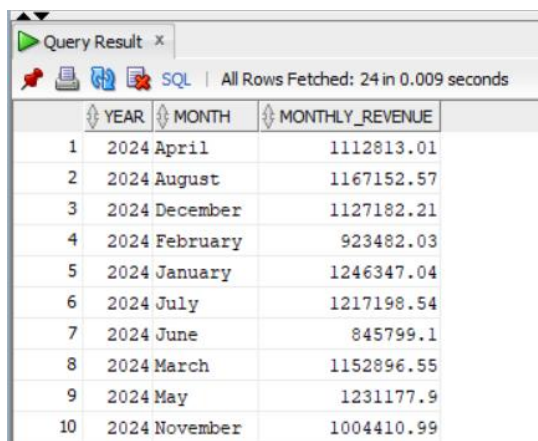
	WAREHOUSE	TOTAL_INVENTORY
1	WH-A	579577

Monthly Revenue Trend

```

SELECT
    YEAR,
    MONTH,
    SUM(REVENUE) AS MONTHLY_REVENUE
FROM SUPPLY_CHAIN
GROUP BY YEAR, MONTH
ORDER BY YEAR, MONTH;

```



Query Result x

All Rows Fetched: 24 in 0.009 seconds

	YEAR	MONTH	MONTHLY_REVENUE
1	2024	April	1112813.01
2	2024	August	1167152.57
3	2024	December	1127182.21
4	2024	February	923482.03
5	2024	January	1246347.04
6	2024	July	1217198.54
7	2024	June	845799.1
8	2024	March	1152896.55
9	2024	May	1231177.9
10	2024	November	1004410.99

Sales Summary View:

```
CREATE VIEW VW_SALES_SUMMARY AS
SELECT
    PRODUCT,
    CATEGORY,
    REGION,
    SUM(SALES_QUANTITY) AS TOTAL_SALES,
    SUM(REVENUE) AS TOTAL_REVENUE
FROM SUPPLY_CHAIN
GROUP BY
    PRODUCT,
    CATEGORY,
    REGION;
```

Monthly Sales View:

```
CREATE VIEW VW_MONTHLY_SALES AS
SELECT
    YEAR,
    MONTH,
    SUM(SALES_QUANTITY) AS MONTHLY_SALES,
    SUM(REVENUE) AS MONTHLY_REVENUE
FROM SUPPLY_CHAIN
GROUP BY
    YEAR,
    MONTH;
```


Region Performance View:

```
CREATE VIEW VW_REGION_PERFORMANCE AS
SELECT
    REGION,
    SUM(SALES_QUANTITY) AS TOTAL_SALES,
    SUM(REVENUE) AS TOTAL_REVENUE,
    AVG(INVENTORY) AS AVG_INVENTORY
FROM SUPPLY_CHAIN
GROUP BY REGION;
```

Inventory Status View:

```
CREATE VIEW VW_INVENTORY_STATUS AS
SELECT
    PRODUCT,
    WAREHOUSE,
    INVENTORY,
    CASE
        WHEN INVENTORY < 100 THEN 'Low Stock'
        WHEN INVENTORY BETWEEN 100 AND 300 THEN 'Medium Stock'
        ELSE 'High Stock'
    END AS STOCK_STATUS
FROM SUPPLY_CHAIN;
```

Supplier Performance View:

```
CREATE VIEW VW_SUPPLIER_PERFORMANCE AS
SELECT
    SUPPLIER,
    AVG(LEAD_TIME_DAYS) AS AVG_LEAD_TIME,
    SUM(REVENUE) AS TOTAL_REVENUE
FROM SUPPLY_CHAIN
GROUP BY SUPPLIER;
```

Demand Summary View:

```
CREATE VIEW VW_DEMAND_SUMMARY AS
SELECT
    PRODUCT,
    AVG(SALES_QUANTITY) AS AVG_DEMAND,
    MAX(SALES_QUANTITY) AS MAX_DEMAND,
    MIN(SALES_QUANTITY) AS MIN_DEMAND
FROM SUPPLY_CHAIN
GROUP BY PRODUCT;
```

Top Products View:

```
CREATE VIEW VW_TOP_PRODUCTS AS
SELECT
    PRODUCT,
    SUM(REVENUE) AS TOTAL_REVENUE,
    DENSE_RANK() OVER(ORDER BY SUM(REVENUE) DESC) AS
PRODUCT_RANK
FROM SUPPLY_CHAIN
GROUP BY PRODUCT;
```

Warehouse Performance View:

```
CREATE VIEW VW_WAREHOUSE_PERFORMANCE AS
SELECT
    WAREHOUSE,
    SUM(SALES_QUANTITY) AS TOTAL_SALES,
    AVG(INVENTORY) AS AVG_INVENTORY
FROM SUPPLY_CHAIN
GROUP BY WAREHOUSE;
```

Product Category View:

```
CREATE VIEW VW_CATEGORY_SUMMARY AS
SELECT
    CATEGORY,
    SUM(REVENUE) AS TOTAL_REVENUE,
    SUM(SALES_QUANTITY) AS TOTAL_SALES
FROM SUPPLY_CHAIN
GROUP BY CATEGORY;
```

Executive Dashboard View:

```
CREATE VIEW VW_EXECUTIVE_DASHBOARD AS
SELECT
    REGION,
    CATEGORY,
    SUM(REVENUE) AS TOTAL_REVENUE,
    SUM(SALES_QUANTITY) AS TOTAL_SALES,
    AVG(INVENTORY) AS AVG_INVENTORY
FROM SUPPLY_CHAIN
GROUP BY
    REGION,
    CATEGORY;
```

Check All Views:

```
SELECT VIEW_NAME FROM USER_VIEWS;
```